

Best Arm Identification in Multi-Armed Bandits with Erasures

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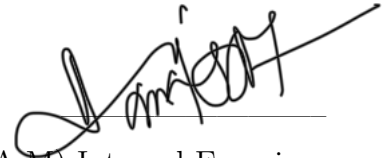
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Abstract

Best arm identification (BAI) is classically studied under the assumption that the learner both chooses which arm is pulled and observes the resulting reward. In many practical systems the first assumption fails: a learner transmits its chosen arm to a far-away agent over an unreliable link, and some of these commands are lost. We study fixed-confidence BAI under such an *erasure channel with finite memory* R . When the learner transmits an arm, with probability $1 - \varepsilon$ it reaches the agent, which pulls it; with probability ε the command is erased and the agent instead repeats a recently pulled arm, resorting to a uniformly drawn arm only after R consecutive erasures. The single parameter R interpolates between two regimes the literature usually treats separately—repeating the previous arm ($R \rightarrow \infty$) and resetting uniformly ($R = 0$). The reward of the arm actually pulled is always observed, but the learner receives no erasure feedback: it never learns which arm was pulled or whether an erasure occurred. It must declare the highest-mean arm with error probability at most δ using as few transmissions as possible.

Two features set this problem apart from classical BAI. First, the learner controls only the transmitted arm, not the arm the agent pulls under an erasure, and it gets no indication that an erasure occurred. Second, because the agent repeats recent arms, the reward observed after transmitting a given arm is drawn from a known-weight *mixture* over the last R transmitted arms. Casting the length- R window of recent transmissions as the state of a controlled Markov chain, we derive an information-theoretic lower bound on the expected sample complexity of any δ -probably-correct policy through a change-of-measure argument: the optimal sampling proportions solve a max–min program over state–action occupancies and the confusing alternatives $\text{Alt}(\mu)$. The bound makes explicit how the erasure probability ε inflates the characteristic time relative to the erasure-free setting, and how it depends on the memory R .

On the achievability side we show the bound is attained, as $\delta \downarrow 0$, by a mixture-KL Track-and-Stop procedure that tracks the lower-bound allocation with forced exploration and stops via a generalized-likelihood-ratio test, evaluating the otherwise intractable mixture KL divergences numerically. For tractable experimentation we also give E-NAS, a reduction that treats the window MDP-with K^R states and a known deterministic shift kernel—as a discounted best-policy-identification problem solved by policy iteration and a convex allocation program. On a two-armed Bernoulli instance ($\mu = [0.30, 0.90]$), E-NAS identifies the correct arm in all 1,100 completed runs, with zero erroneous declarations; its mean stopping time grows linearly in $\log(1/\delta)$

across confidence levels from e^{-1} down to e^{-15} , matching the lower-bound rate, and decreases monotonically as the erasure probability is reduced—directly quantifying the price of erasures.

More broadly, this work isolates the fundamental price erasures impose on pure exploration: sample complexity is governed not by the raw number of requests but by the rate at which un-erased, informative observations accumulate under a memory-limited agent, at a computational cost that grows with the window as K^R . We close by outlining extensions—feeding erasure indicators back to the learner, the infinite-memory limit $R \rightarrow \infty$, non-asymptotic guarantees, and adaptivity to an unknown ε or R -toward best arm identification over genuinely unreliable feedback channels.

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Chapter 1

Introduction

1.1 Background and Motivation

1.1.1 Multi-Armed Bandits

The multi-armed bandit (MAB) is the canonical model of sequential decision-making under uncertainty. A learner is faced with a finite set of K *arms*, each associated with an unknown reward distribution. At every round the learner selects one arm, or *pulls* it, and receives a reward drawn from that arm’s distribution. Because the learner observes only the arms it chooses, every decision trades off acquiring information about poorly understood arms against exploiting arms that already look good. This exploration–exploitation tension, formalized in the seminal work of Lai and Robbins [LR85], has made the bandit framework a foundational tool across clinical trials, online recommendation, communication networks, and adaptive experimentation [LS20].

1.1.2 Best Arm Identification and Pure Exploration

In many applications the eventual goal is not to maximize reward *during* the experiment but to use a budget of samples to reach a confident *conclusion*—for example, to recommend the single best treatment at the end of a trial. This is the *best arm identification* (BAI) problem, the standard and most representative instance of pure exploration [BMS11]. Here the learner samples arms adaptively and, at a stopping time of its choosing, declares which arm it believes has the highest mean. The performance criterion is no longer cumulative regret but *sample complexity*: how many samples are required to identify the best arm with a prescribed level of confidence.

A long line of work has characterized the information-theoretic limits of this problem and designed algorithms that meet them [EDMM06, KCG16, GK16].

1.2 The Challenge of Unreliable Feedback

1.2.1 Sources of Erasures

Classical BAI rests on an assumption: the learner both decides which arm to pull and pulls it. In many real deployments these two roles are separated. The entity that decides-the *learner*-is remote from the entity that acts in the environment-the *agent*-and communicates its decisions over an unreliable link. A controller dispatching commands to a low-power field device, a base station instructing a sensor, or any setup in which actions are signalled over a noisy channel fits this picture: commands are dropped, transmissions time out, and a fraction of the learner’s chosen arms simply never reach the agent. Borrowing the terminology from the communications field, we call such a lost command an *erasure*.

1.2.2 Impact on Learning

Erasures do more than slow down data collection. When the agent acting in the environment is resource-limited, it cannot buffer an unbounded history or wait indefinitely for an acknowledgement before acting again. Its behaviour upon an erasure is therefore governed by a simple, hard-wired rule rather than by a deliberate decision of the learner. As a result the data stream the learner observes is no longer a clean sequence of independent pulls of chosen arms: it is filtered through the lossy channel with memory. Understanding how much this degrades statistical efficiency and how to act optimally in spite of it is the central question of this thesis.

1.3 Learning Objectives and Feedback Models

1.3.1 Regret Minimization versus Pure Exploration

Two objectives dominate the bandit literature (Figure 1.1). In *regret minimization* the learner seeks to maximize the total reward accumulated over time, which forces it to keep exploiting near-optimal arms. In *pure exploration*, and BAI in particular, intermediate rewards are irrelevant; only the quality of the final recommendation matters. This thesis is concerned exclusively with the latter, because the motivating

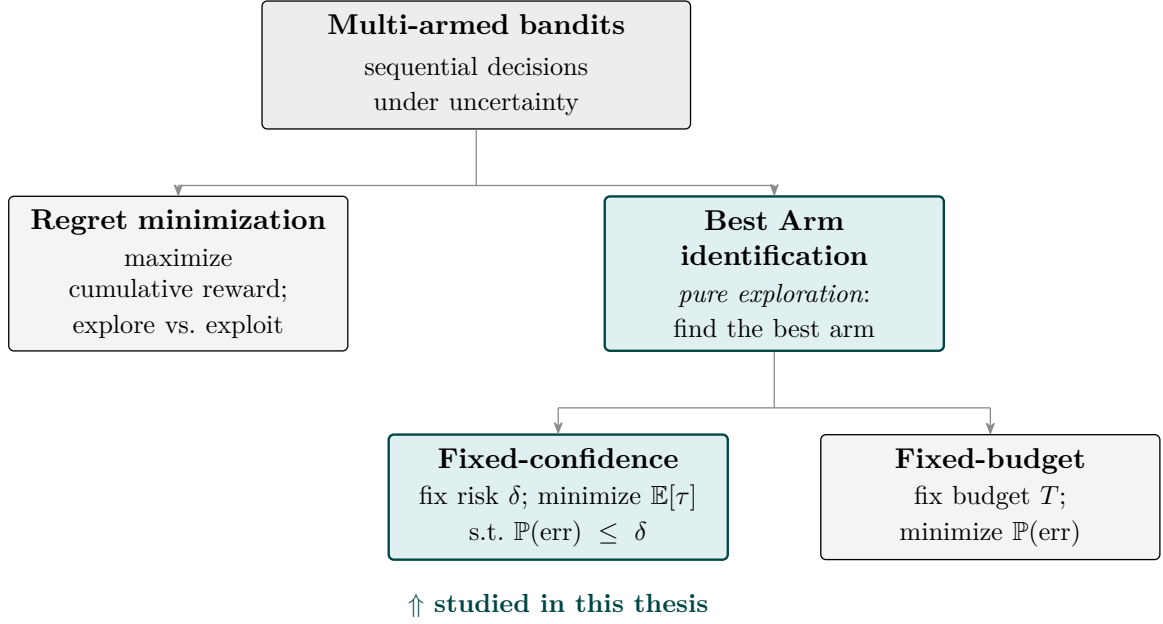


Figure 1.1: Where this thesis sits in the bandit landscape. By *objective*, bandit problems split into regret minimization and best arm identification—the prototypical pure-exploration problem; BAI in turn comes in *fixed-confidence* and *fixed-budget* formulations, which are not in general reducible to one another. This thesis adopts the highlighted path, fixed-confidence BAI, because it admits the instance-optimal change-of-measure lower bound on which the rest of the thesis is built.

applications—selecting the best configuration, channel, or treatment under unreliable feedback—are naturally identification problems.

1.3.2 Fixed-Confidence and Fixed-Budget Formulations

BAI comes in two complementary flavours (Figure 1.1). In the *fixed-budget* setting the number of samples is fixed in advance and the goal is to minimize the probability of an incorrect recommendation. In the *fixed-confidence* setting, which we adopt, the learner is given a target error probability δ and must guarantee that it declares the wrong arm with probability at most δ , while using as few samples as possible. The fixed-confidence formulation admits a clean information-theoretic lower bound via change-of-measure arguments [KCG16], and a matching family of *Track-and-Stop* algorithms [GK16], which makes it the natural lens through which to study the price of erasures.

1.4 Limitations of Classical BAI under Erasures

The asymptotically optimal algorithms for classical BAI assume that the learner both *chooses* which arm is pulled and *observes* the resulting reward. Neither holds in our setting. First, when a transmission is erased, the sample that arrives may not correspond to the arm the learner intended, because the agent falls back on a fixed rule. Second, the learner never sees which arm actually produced a reward, nor whether an erasure occurred; it sees only the reward value. Applying off-the-shelf BAI algorithms naively-treating each received reward as a clean pull of the requested arm-therefore biases the estimates and voids their correctness guarantees. A principled treatment must model the channel explicitly and re-derive both the fundamental limits and the algorithms.

1.5 The Agent–Learner Architecture

1.5.1 The Learner and the Agent

We make the separation between deciding and acting explicit (Figure 1.2). The *learner* performs all statistical work: at each round it selects an arm, transmits it to the agent, estimates the arm means from the rewards it observes, decides when enough evidence has accumulated, and issues the final recommendation. The *agent* merely executes: when it receives the transmitted arm it pulls that arm, and when the transmission is erased it falls back on a fixed, causal rule. The agent is a simple, resource-limited device rather than a second learner, so this fallback is hard-wired rather than adaptively optimized-which is precisely why the otherwise unusual “simple” behaviours studied here are the realistic ones. The reward of the arm the agent pulls is always reported back to the learner.

1.5.2 The Finite-Memory Erasure Channel

Concretely, we model the learner-to-agent link as an *erasure channel with finite memory* R . When the learner transmits an arm, with probability $1 - \varepsilon$ it reaches the agent, which pulls it. With probability ε the transmitted arm is erased; the agent then repeats the arm it last pulled, and only after R consecutive erasures does it reset by pulling an arm uniformly at random. The single parameter R thus interpolates between two regimes that the literature usually treats as separate special cases: always repeating the previous arm ($R \rightarrow \infty$) and resetting uniformly ($R = 0$). The reward

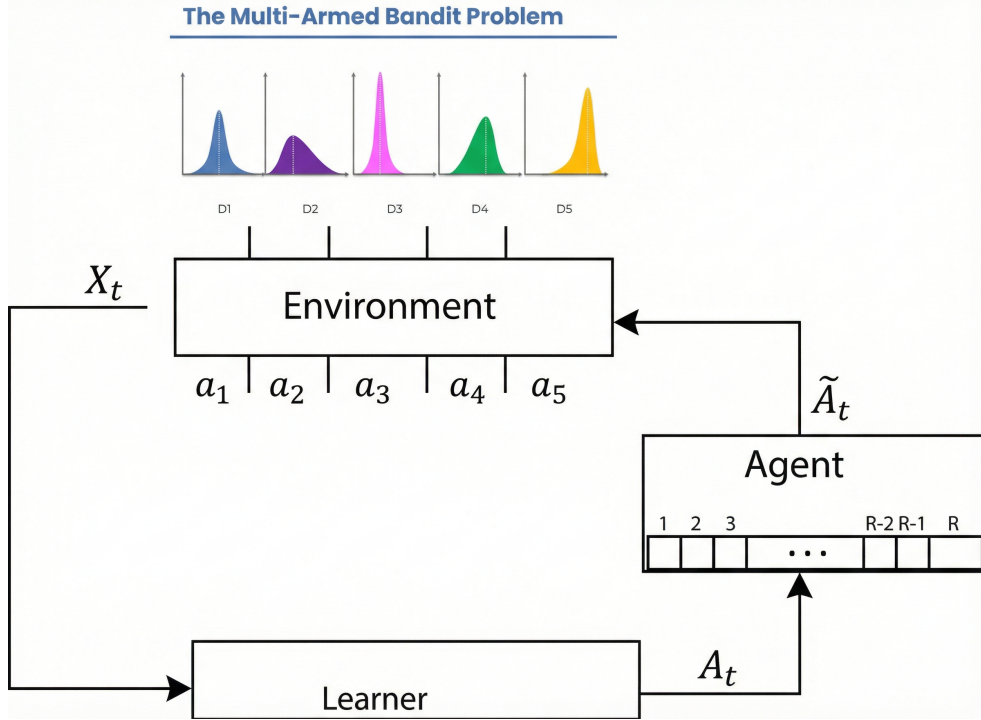


Figure 1.2: The best arm identification problem over a finite-memory erasure channel. The learner transmits a chosen arm A_t to the agent. With probability $1 - \varepsilon$ the command arrives and the agent pulls it, $\tilde{A}_t = A_t$; with probability ε it is erased and the agent instead repeats a recently pulled arm held in its length- R memory, resetting to a uniformly random pull only after R consecutive erasures. The environment, a bank of K arms with reward distributions $D_1, \dots, D_K$, returns the reward X_t of the pulled arm $\tilde{A}_t$ to the learner, which never observes $\tilde{A}_t$ nor whether an erasure occurred.

of the pulled arm is always observed, but the learner is given no erasure feedback: it never learns the pulled arm or whether an erasure occurred.

1.6 Problem Statement

1.6.1 Core Challenge

Given a K -armed bandit observed through a finite-memory erasure channel with known parameters ε and R , and a target confidence $\delta \in (0, 1)$, we seek a policy—an arm-selection rule, a stopping rule, and a recommendation rule—that declares the highest-mean arm with error probability at most δ while minimizing the expected number of requests. The difficulty is that each received reward is a sample from a

mixture of arm distributions induced by the channel’s memory, so the learner must invert this mixing to recover the underlying arm means.

1.6.2 Research Gap

Erasures and lossy feedback have been studied in the context of regret minimization, and pure exploration has been studied extensively under clean feedback, but the intersection-fixed-confidence BAI under a memory-bearing erasure channel with a resource-constrained agent has received little attention. In particular, the fundamental sample-complexity limit of this problem and an algorithm that attains it have not, to our knowledge, been characterized. This thesis addresses that gap.

1.7 Proposed Approach

1.7.1 Design Intuition

Our approach rests on a single modelling insight: although the learner cannot see the realized arm, the channel’s behaviour is fully described by its parameters ε and R together with the agent’s length- R request window. We can therefore treat the window as the state of a controlled Markov chain, express the reward of each request as a *known-weight* mixture over the arms in that window, and recover the classical change-of-measure machinery in this lifted space. Optimality is then a statement about how to allocate requests across window states so as to distinguish the best arm from its closest competitors as quickly as possible.

1.7.2 E-NaS: A Tractable MDP Reduction

To make large-scale experimentation feasible we also develop E-NaS, a reduction that casts the length- R window as the state of a discounted Markov decision process with K^R states and a known deterministic shift kernel, and solves the resulting *best-policy-identification* problem with the adaptive-sampling machinery of Al Marjani and Proutiere [AMP21, AMGP21]: policy iteration to evaluate the current best policy, a convex program to set the allocation, forced-exploration tracking, and a GLR stopping rule. E-NaS is the algorithm used in all experiments reported in this thesis.

1.8 Key Contributions

The contributions of this thesis are as follows.

- We formalize fixed-confidence best arm identification over a finite-memory erasure channel, making explicit the separation between a resource-constrained agent and the decision-making learner, and showing that the single memory parameter R unifies the uniform-reset and always-repeat behaviours studied separately in prior work (Chapters 2 and 3).
- We derive an information-theoretic lower bound on the expected sample complexity of any δ -probably-correct policy via a change-of-measure argument, expressed as a max–min program over state–action occupancies, and make explicit how the erasure probability ε and the memory R enter the characteristic time (Chapter 4).
- We propose an algorithm: a tractable reduction to discounted best-policy identification on the window MDP (Chapter 4).
- We provide an extensive empirical study on Bernoulli bandit instances, confirming correct identification across all completed runs and showing that the empirical stopping time scales linearly in $\log(1/\delta)$ and grows with the erasure probability, directly quantifying the price of erasures (Chapter 5); the complete implementation is released openly at <https://github.com/Deekshithrathod/Erasure-BAI>.
- We give a geometric and information-theoretic interpretation of why erasures inflate sample complexity, and chart concrete extensions toward BAI over genuinely unreliable feedback channels (Chapters 6 and 7).

Thesis organization. Chapter 2 surveys the relevant literature. Chapter 3 sets up the bandit model, the finite-memory erasure channel, and the MDP formulation. Chapter 4 develops the lower bound and the E-NAS algorithm. Chapter 5 reports the experimental evaluation. Chapter 6 interprets the results, and Chapter 7 concludes with directions for future work.

Chapter 2

Literature Review

This chapter situates the thesis within four bodies of work: sequential hypothesis testing, classical best arm identification, the tracking and elimination algorithms that attain its limits, and the literature on learning under lossy or erased feedback. We close by identifying the gap that the thesis addresses.

2.1 Sequential Hypothesis Testing

The roots of fixed-confidence identification lie in sequential analysis. Wald’s sequential probability ratio test [Wal45] established that, when samples may be collected adaptively, a target error level can be met with far fewer observations than a fixed-sample test requires. Chernoff [Che59] extended this to the active setting, in which the experimenter also chooses *which* experiment to run at each step. best arm identification is precisely an active sequential test: the “experiments” are arm pulls and the competing hypotheses are “arm a is best.” The generalized-likelihood-ratio statistic and the $\log(1/\delta)$ scaling of stopping times that recur throughout this thesis are direct descendants of these ideas.

2.2 Classical best arm Identification

2.2.1 Fixed-Confidence Formulation

In the fixed-confidence setting the learner fixes a risk level δ and minimizes the expected number of samples subject to a δ -correctness guarantee. Kaufmann, Cappé and Garivier [KCG16] characterized the instance-dependent sample-complexity lower bound through a change-of-measure argument, showing that the optimal asymptotic

sample complexity is $T^*(\mu) \log(1/\delta)$, where the *characteristic time* $T^*(\mu)$ is the value of a max–min optimization over sampling proportions and confusing alternative instances. This bound is the template we lift to the erasure setting in Chapter 4.

2.2.2 Fixed-Budget Formulation

The complementary fixed-budget setting [BMS11] fixes the number of samples and minimizes the error probability. Although we do not pursue it, it provides useful contrast: the two formulations are not in general reducible to one another, and the algorithmic techniques differ. We adopt the fixed-confidence view because it yields the clean lower bound and matching algorithm that let us quantify the price of erasures.

2.3 Algorithmic Paradigms for Fixed-Confidence BAI

2.3.1 Successive Elimination

Action-elimination methods maintain confidence intervals on the arm means and discard arms whose intervals fall below that of an empirically better arm [EDMM02, EDMM06]. They are simple and robust and achieve near-optimal sample complexity up to logarithmic factors, but they are not in general asymptotically optimal because their uniform exploration does not match the instance-optimal allocation.

2.3.2 Tracking-Based Methods

Garivier and Kaufmann [GK16] introduced *Track-and-Stop*, which closes the gap to the lower bound. The algorithm repeatedly (i) re-estimates the means, (ii) computes the optimal allocation that solves the characteristic-time program at the current estimate, (iii) *tracks* that allocation with forced exploration so that every arm is sampled infinitely often, and (iv) stops when a GLR statistic crosses a $\log(1/\delta)$ -type threshold. Our asymptotically optimal target (Chapter 4) adapts this paradigm to the mixture observations induced by the channel.

2.4 Best-Policy Identification in MDPs

When the quantity to be identified is the optimal policy of a Markov decision process rather than the best of K arms, the same change-of-measure philosophy applies but the geometry is richer. Al Marjani and Proutiere [AMP21] and Al Marjani, Garivier and Proutiere [AMGP21] developed adaptive-sampling algorithms—collectively the “MDP Navigate-and-Stop” family—that compute an allocation over state–action pairs subject to navigation (flow) constraints and track it through the controlled dynamics. Because our finite-memory channel turns the BAI problem into a structured MDP over request windows, this line of work is the direct foundation for our tractable algorithm E-NAS.

2.5 Erasures and Lossy Feedback in Bandits

Unreliable feedback has been considered mainly in the regret-minimization literature, where observations may be delayed, anonymized, aggregated, or dropped. These works typically assume the learner still controls which arm is pulled and that losses are independent of the arm history. Our setting differs in two ways that matter: the loss process has *memory* (the channel repeats arms and resets only after R erasures), and the acting agent is decoupled from the learner and resource-constrained. To our knowledge the fixed-confidence, pure-exploration counterpart of this problem has not been analyzed.

2.6 Gap in the Literature

In summary, fixed-confidence BAI is well understood under clean feedback, lossy feedback has been studied chiefly for regret, and MDP best-policy identification provides the right machinery for structured exploration—but no existing work characterizes the fundamental limit of pure exploration under a memory-bearing erasure channel served by a constrained agent, nor provides an algorithm that attains it. Closing this gap is the goal of the remaining chapters.

Chapter 3

Theoretical Foundations

This chapter formalizes the problem. We introduce the bandit model, define the finite-memory erasure channel and the mixture distributions it induces, state the identification objective, and recast the problem as a Markov decision process over request windows. The change-of-measure preliminaries collected at the end are used to derive the lower bound in Chapter 4.

3.1 Notation and the Bandit Model

Let $[K] = \{1, \dots, K\}$ index the arms. Each arm $a \in [K]$ has an unknown reward distribution ν_a with mean μ_a ; write $\mu = (\mu_1, \dots, \mu_K)$ and $\nu = (\nu_1, \dots, \nu_K)$. We make no parametric assumption on the ν_a : the lower bound and algorithms of Chapter 4 are expressed entirely through Kullback–Leibler divergences between the distributions the channel induces, and apply to any reward distributions for which these divergences are well defined. So that an instance is summarized by its mean vector μ and the confusing alternatives can be indexed by competing mean vectors, we take the ν_a to belong to a common family identified by its mean (for instance a single-parameter exponential family). The bounded, Bernoulli instantiation is introduced only for the experiments of Chapter 5, where the best-policy-identification reduction underlying E-NAS requires bounded rewards.

Assumption 3.1 (Unique best arm). *There is a unique optimal arm $a^*(\mu) = \arg \max_{a \in [K]} \mu_a$.*

The learner’s goal is to identify $a^*(\mu)$. It selects arms and transmits them to an agent over the erasure channel defined next; it always observes the resulting reward, but on an erasure it does not learn which arm the agent actually pulled.

3.2 The Finite-Memory Erasure Channel

3.2.1 Channel Dynamics

At round n the learner *selects* an arm $A_n \in [K]$ and transmits it to the agent over the erasure channel. Independently across rounds, with probability $1 - \varepsilon$ the transmission succeeds and the agent pulls the received arm, $\tilde{A}_n = A_n$. With probability ε the transmitted arm is erased and never reaches the agent; the agent, aware that an erasure has occurred, instead pulls an arm according to a fixed causal rule. We consider three such rules of how an agent should act after erasure event: (i) pull an arm uniformly at random; (ii) for a fixed $R \in \mathbb{N}$, repeat the previously pulled arm while at most R consecutive erasures have occurred and pull uniformly at random on the $(R+1)$ st consecutive erasure; and (iii) always repeat the previously pulled arm. Rule (i) is the case $R = 0$ of rule (ii), and rule (iii) is its limit $R \rightarrow \infty$. The reward X_n generated by the pulled arm $\tilde{A}_n$ is observed by both the agent and the learner. Crucially, the learner receives *no erasure feedback*: it observes X_n but never learns $\tilde{A}_n$ or whether an erasure occurred. The learner knows ε and the agent's rule.

3.2.2 Effective Reward Distributions

The key consequence of these dynamics is that, conditioned on the recent history of transmitted arms, each observed reward is drawn from a *known-weight mixture* of the arm distributions. Encode the recent history by the window of the last R transmitted arms, $s = (a_1, \dots, a_R) \in [K]^R$, and let $b \in [K]$ be the arm transmitted at the current round. Under rule (ii) the observation is distributed as

$$\nu_{s,b,\varepsilon}^\mu = (1 - \varepsilon) \nu_b + (1 - \varepsilon) \sum_{j=1}^R \varepsilon^{R+1-j} \nu_{a_j} + \frac{\varepsilon^{R+1}}{K} \sum_{b'=1}^K \nu_{b'}. \quad (3.1)$$

whose mixture weights sum to one and depend only on the known parameters ε and R . We write $\theta_{s,b,\varepsilon}$ for the mean of $\nu_{s,b,\varepsilon}^\mu$, i.e. the same convex combination of the entries of μ . The mixing in Equation (3.1) is the central statistical obstacle: the learner must infer μ from samples of these mixtures, and—as noted below—the KL divergence between two such mixtures has no closed form.

3.2.3 Special Cases

The single parameter R recovers the two regimes treated separately in prior work. When $R = 0$, every erasure triggers an immediate uniform pull, recovering rule (i); the effective distribution collapses to $\nu_{b,\varepsilon,\text{unif}}^\mu = (1 - \varepsilon)\nu_b + \frac{\varepsilon}{K} \sum_{a'} \nu_{a'}$. As $R \rightarrow \infty$ the uniform-reset term vanishes and the agent always repeats the previous arm on an erasure, recovering rule (iii). Intermediate R interpolates between these extremes, which is why a single analysis covers the “uniform,” “ R -tolerant,” and “always-repeat” behaviours.

3.2.4 The Memory Induced by Repetition

It is worth making explicit the memory that the repeat rule injects into the observations, since it is the feature that distinguishes this problem from classical best arm identification. Conditioned on the transmitted history, the mean of the reward observed under the R -tolerant rule is, for $t \geq R$,

$$\mathbb{E}_\mu[X_t \mid A_{0:t}, X_{0:t-1}] = (1 - \varepsilon) \sum_{s=0}^R \varepsilon^s \mu_{A_{t-s}} + \frac{\varepsilon^{R+1}}{K} \sum_{a'=1}^K \mu_{a'}, \quad (3.2)$$

the mean $\theta_{s,b,\varepsilon}$ of the mixture (3.1) written in terms of the last $R+1$ transmitted arms $A_t, \dots, A_{t-R}$ (the derivation is in Appendix A.1). Two limits bracket this expression. At $R = 0$ the sum collapses to $(1 - \varepsilon)\mu_{A_t} + \frac{\varepsilon}{K} \sum_{a'} \mu_{a'}$, which depends on the current arm A_t alone: uniform reset is *memoryless*. As $R \rightarrow \infty$ the reset term vanishes and the conditional mean becomes

$$\mathbb{E}_\mu[X_t \mid A_{0:t}, X_{0:t-1}] = (1 - \varepsilon) \sum_{u=0}^t \varepsilon^{t-u} \mu_{A_u} + \frac{\varepsilon^{t+1}}{K} \sum_{a'=1}^K \mu_{a'},$$

a function of the *entire* selection history $A_{0:t}$: always-repeat has unbounded memory. The parameter R is therefore a literal memory horizon, and the agent’s behaviour ranges over a spectrum from memoryless ($R = 0$) to full-history ($R \rightarrow \infty$).

This memory is what makes the always-repeat rule the analytically hardest of the three. Because its conditional law depends on every past selection, the sample complexity of an optimal δ -PC policy does not reduce to a finite-dimensional optimization, and characterizing it exactly remains open. Truncating the memory at a finite R restores a finite sufficient statistic, the length- R window, which is what the analysis below relies on.

3.2.5 Why Finite Memory?

Two features of this model are dictated by the agent being resource-constrained, and each corresponds to a single ingredient. The agent's *memory* budget is exactly the parameter R : it can retain only the last R pulled arms, so under a run of erasures it can do no more than repeat from that bounded buffer, and once the buffer is exhausted after R consecutive losses it must take the one action that needs no memory at all, a uniform pull. The agent's *compute* budget is what forces the fallback to be a fixed, causal, hard-wired rule rather than an adaptive decision: solving an optimization on each loss is the learner's job, not the agent's. The rules (i)–(iii) are therefore not arbitrary but the minimal-resource behaviours of a single memory budget, with uniform reset ($R = 0$) and always-repeat ($R \rightarrow \infty$) as its two extremes.

The same finite-memory assumption is also what makes the problem tractable. Because the agent looks back at most R steps, the length- R window of recent transmissions is a *sufficient statistic* for the law of the next reward. This is precisely what renders each observation a known-weight mixture (3.1) and lets the next section recast the dynamics as a controlled Markov chain with a known, deterministic kernel, on which the change-of-measure lower bound and tracking algorithms of Chapter 4 are built. An unbounded, adaptive agent would instead induce an uncontrolled, infinite-memory observation process with no comparable structure. Nothing below uses the specific form of the fallback beyond its being causal and depending on bounded memory, so the analysis carries over unchanged to other such rules, with only the mixture weights and the shift kernel changing.

3.3 Policies and the Identification Objective

3.3.1 Arm-Selection, Stopping, and Recommendation Rules

A policy $\pi = ((A_n)_{n \geq 0}, \tau, \hat{a})$ is run by the learner and consists of three components: a *sampling rule* that selects the arm A_n to transmit, based on the observed history $(A_{0:n-1}, X_{0:n-1})$; a *stopping rule* τ that decides when to halt; and a *recommendation rule* $\hat{a} \in [K]$ that outputs the estimated best arm at time τ . The agent's erasure fallback rule is fixed and known to the learner, and is not itself part of the policy. The length- R window of recent selections enters only as a convenient sufficient statistic for the R -tolerant analysis below.

3.3.2 δ -Probably-Correct Policies and Sample Complexity

definition

[. δ -PC policy] For $\delta \in (0, 1)$, a policy is δ -probably-correct on a class of instances if, for every instance μ in the class, $\mathbb{P}_\mu(\tau < \infty, \hat{a} \neq a^*(\mu)) \leq \delta$.

Among δ -PC policies we seek to minimize the expected sample complexity $\mathbb{E}_\mu[\tau]$. The remainder of the thesis characterizes the smallest achievable $\mathbb{E}_\mu[\tau]$ and constructs policies that attain it.

3.4 A Markov Decision Process over Request Windows

3.4.1 State Space, Actions, and the Shift Kernel

Because the mixture (3.1) depends on the history only through the length- R window, the window is a sufficient state. Define the state space $\mathcal{S} = [K]^R$, so $|\mathcal{S}| = K^R$, and let the action be the requested arm $b \in [K]$. The window evolves deterministically by a shift,

$$s' = \text{Shift}(s, b) = (a_2, \dots, a_R, b), \quad P(s' \mid s, b) = \mathbf{1}\{s' = \text{Shift}(s, b)\}. \quad (3.3)$$

This yields a controlled Markov chain whose kernel is *known and deterministic*; all stochasticity is confined to the rewards through the mixture in (3.1).

3.4.2 Rewards and the Best-Policy Correspondence

Assigning to each state-action pair the mixture mean $\theta_{s,b,\varepsilon}$ as its reward turns the window chain into a Markov decision process. Identifying the best arm is then equivalent to identifying an optimal policy of this MDP: under Assumption 3.1, the action that maximizes long-run reward from every state is to request $a^*(\mu)$. This equivalence is what lets us import the best-policy-identification machinery of [AMP21, AMGP21] in Chapter 4; a discount factor $\gamma \in (0, 1)$ is used there as an algorithmic device to make the value functions well defined.

3.5 Change-of-Measure Preliminaries

The lower bound rests on a standard transportation argument. For an instance μ , the set of *confusing alternatives* is

$$\text{Alt}(\mu) = \{\lambda \in \mathbb{R}^K : a^*(\lambda) \neq a^*(\mu)\}, \quad (3.4)$$

the instances under which a different arm is optimal. A δ -PC policy must gather enough evidence to separate μ from every $\lambda \in \text{Alt}(\mu)$; by the data-processing / transportation inequality this evidence is measured in expected log-likelihood ratio, i.e. in Kullback–Leibler divergence between the observation laws under μ and under λ [KCG16, CT06]. In our setting the relevant divergences are between the effective distributions $\nu_{s,b,\varepsilon}^\mu$ and $\nu_{s,b,\varepsilon}^\lambda$ of (3.1). The expected log-likelihood ratio decomposes exactly into a weighted sum of these divergences over state–action visit counts (Chapter 4), which is what makes the change-of-measure lower bound tractable. In the special case of uniform fallback ($R = 0$) the analysis reduces further to classical best arm identification on the effective distributions $\nu_{a,\varepsilon,\text{unif}}$, whose means are the shifted means $\mu_a^\varepsilon = (1 - \varepsilon)\mu_a + \frac{\varepsilon}{K} \sum_{a'} \mu_{a'}$.

Chapter 4

Proposed Methodology

This chapter develops the theory. We first compute the expected log-likelihood ratio (LLR) of the learner's observations and identify the effective reward distributions induced by the channel. Using these, we prove information-theoretic lower bounds on the expected sample complexity of any δ -PC policy via a change-of-measure argument. We then present a tracking-based algorithm whose stopping time matches the lower bound as $\delta \downarrow 0$, and describe E-NAS, the tractable Markov-decision-process reduction used in all experiments. Throughout, recall the alternative set $\text{Alt}(\mu)$ of (3.4), and write $d(x, y) := x \log \frac{x}{y} + (1 - x) \log \frac{1-x}{1-y}$ for the binary relative entropy. The bounds below involve only Kullback–Leibler divergences between the effective distributions the channel induces and assume no parametric form for the rewards.

4.1 Log-Likelihood Ratios and Effective Distributions

Fix a policy π and two instances $\mu, \lambda \in \mathbb{R}^K$. The LLR of the arm selections and rewards observed up to time n is

$$Z_{\mu, \lambda}(n) = \log \frac{\mathbb{P}_{\mu}(A_{0:n}, X_{0:n})}{\mathbb{P}_{\lambda}(A_{0:n}, X_{0:n})} = \sum_{t=0}^n \log \frac{\mathbb{P}_{\mu}(X_t \mid A_{0:t}, X_{0:t-1})}{\mathbb{P}_{\lambda}(X_t \mid A_{0:t}, X_{0:t-1})}, \quad (4.1)$$

where the arm-selection terms cancel because the sampling rule is oblivious to the instance. The conditional law of X_t depends on the agent's fallback rule.

Uniform fallback ($R = 0$). When the agent samples uniformly under erasures, for every $t \geq 0$,

$$\mathbb{P}_\mu(X_t \mid A_{0:t}, X_{0:t-1}) = (1 - \varepsilon) \nu_{A_t}(X_t) + \frac{\varepsilon}{K} \sum_{a'=1}^K \nu_{a'}(X_t) =: \nu_{A_t, \varepsilon, \text{unif}}(X_t), \quad (4.2)$$

with ν_a the reward distribution of arm a . Taking expectations in (4.1) and using the tower property gives

$$\mathbb{E}_\mu[Z_{\mu, \lambda}(n)] = \sum_{a=1}^K \mathbb{E}_\mu[N_a(n)] \text{KL}(\nu_{a, \varepsilon, \text{unif}} \parallel \nu'_{a, \varepsilon, \text{unif}}), \quad (4.3)$$

where $N_a(n) = \sum_{t=0}^n \mathbf{1}\{A_t = a\}$ counts selections of arm a and ν' denotes the corresponding quantity under λ .

R -tolerant fallback. When the agent tolerates up to R consecutive erasures, unrolling the recursion $\tilde{A}_t = A_t$ (no erasure) or $\tilde{A}_t = \tilde{A}_{t-1}$ (erasure) for $t \geq R$ yields

$$\mathbb{P}_\mu(X_t \mid A_{0:t}, X_{0:t-1}) = (1 - \varepsilon) \sum_{s=0}^R \varepsilon^s \nu_{A_{t-s}}(X_t) + \frac{\varepsilon^{R+1}}{K} \sum_{a'=1}^K \nu_{a'}(X_t) = \nu_{s(t), A_t, \varepsilon}(X_t), \quad (4.4)$$

where $s(t) = (A_{t-R}, \dots, A_{t-1})$ is the window of the last R selections and $\nu_{s, b, \varepsilon}$ is the mixture of (3.1). As noted in Chapter 3, $\{s(t)\}_{t \geq R}$ is a controlled Markov chain with the deterministic shift kernel (3.3). Taking expectations,

$$\mathbb{E}_\mu[Z_{\mu, \lambda}(n)] = \underbrace{\mathbb{E}_\mu \left[\sum_{t=0}^{R-1} \text{KL}(\nu_{s(t), \varepsilon} \parallel \nu'_{s(t), \varepsilon}) \right]}_{\text{boundary terms, } O(1)} + \sum_{s \in [K]^R} \sum_{b=1}^K \mathbb{E}_\mu[N_{s, b}(n)] \text{KL}(\nu_{s, b, \varepsilon} \parallel \nu'_{s, b, \varepsilon}), \quad (4.5)$$

where $N_{s, b}(n) = \sum_{t=R}^n \mathbf{1}\{s(t) = s, A_t = b\}$ counts visits to the state-action pair (s, b) . The R boundary terms are bounded uniformly in n and play no role in the asymptotics below.

4.2 Information-Theoretic Lower Bounds

The converse rests on the transportation lemma of Kaufmann, Cappé and Garivier [KCG16].

Lemma 4.1 (Transportation inequality). *For any δ -PC policy and any $\lambda \in \text{Alt}(\mu)$, $\mathbb{E}_\mu[Z_{\mu,\lambda}(\tau)] \geq d(\delta, 1 - \delta)$.*

The proof is the standard data-processing argument applied to the event $\{\hat{a} \neq a^*(\lambda)\}$, whose probability is at most δ under μ and at least $1 - \delta$ under λ ; we refer to [KCG16, Lemma 19], and give a self-contained proof in Appendix A.3.

Proposition 4.2 (Lower bound, uniform fallback). *For any instance μ with a unique best arm,*

$$\liminf_{\delta \downarrow 0} \inf_{\pi \in \Pi(\delta)} \frac{\mathbb{E}_\mu[\tau]}{\log(1/\delta)} \geq \frac{1}{T_{\varepsilon, \text{unif}}^*(\mu)}, \quad T_{\varepsilon, \text{unif}}^*(\mu) := \sup_{w \in \Sigma_K} \inf_{\lambda \in \text{Alt}(\mu)} \sum_{a=1}^K w_a \text{KL}(\nu_{a, \varepsilon, \text{unif}} \| \nu'_{a, \varepsilon, \text{unif}}),$$

where Σ_K is the simplex over $[K]$.

Proof. Fix a δ -PC policy with $\mathbb{E}_\mu[\tau] < \infty$ (else the bound is trivial). Combining Lemma 4.1 with the identity (4.3) evaluated at the stopping time τ , for every $\lambda \in \text{Alt}(\mu)$,

$$d(\delta, 1 - \delta) \leq \sum_{a=1}^K \mathbb{E}_\mu[N_a(\tau)] \text{KL}(\nu_{a, \varepsilon, \text{unif}} \| \nu'_{a, \varepsilon, \text{unif}}).$$

Write $w_a = \mathbb{E}_\mu[N_a(\tau)] / \mathbb{E}_\mu[\tau + 1] \in \Sigma_K$ (since $\sum_a N_a(\tau) = \tau + 1$). Taking the infimum over λ and then the supremum over the induced w ,

$$d(\delta, 1 - \delta) \leq \mathbb{E}_\mu[\tau + 1] \sup_{w \in \Sigma_K} \inf_{\lambda \in \text{Alt}(\mu)} \sum_a w_a \text{KL}(\nu_{a, \varepsilon, \text{unif}} \| \nu'_{a, \varepsilon, \text{unif}}) = \mathbb{E}_\mu[\tau + 1] T_{\varepsilon, \text{unif}}^*(\mu).$$

Dividing by $\log(1/\delta)$ and using $d(\delta, 1 - \delta) / \log(1/\delta) \rightarrow 1$ as $\delta \downarrow 0$ gives the claim. $\square$

Remark 4.1 (Reduction to classical BAI). Under uniform fallback the law (4.2) of the observed reward when arm a is selected is the mixture $\nu_{a, \varepsilon, \text{unif}}$ with mean $\mu_a^\varepsilon = (1 - \varepsilon)\mu_a + \frac{\varepsilon}{K} \sum_{a'} \mu_{a'}$. Because this mean is an order-preserving affine transform of μ , $a^*(\mu^\varepsilon) = a^*(\mu)$. Identifying the best arm under erasures is thus equivalent to classical best arm identification on the effective distributions $\{\nu_{a, \varepsilon, \text{unif}}\}_a$; the characteristic time above coincides with the classical one of [GK16] applied to these distributions. This is the basis of the successive-elimination guarantee in Section 4.5.

Proposition 4.3 (Lower bound, R -tolerant fallback). *Fix $R \in \mathbb{N}$. For any instance μ with a unique best arm,*

$$\liminf_{\delta \downarrow 0} \inf_{\pi \in \Pi(\delta)} \frac{\mathbb{E}_\mu[\tau]}{\log(1/\delta)} \geq \frac{1}{T_{\varepsilon, R}^*(\mu)},$$

where, with $\nu_{s,b,\varepsilon}$ as in (3.1),

$$T_{\varepsilon,R}^*(\mu) := \sup_{\nu \in \Sigma([K]^{R+1})} \inf_{\lambda \in \text{Alt}(\mu)} \sum_{s \in [K]^R} \sum_{b=1}^K \nu(s, b) \text{KL}(\nu_{s,b,\varepsilon} \| \nu'_{s,b,\varepsilon}),$$

the supremum being over all distributions on state-action pairs.

Proof. The argument mirrors Proposition 4.2, now over state-action pairs. By Lemma 4.1 and (4.5), for every $\lambda \in \text{Alt}(\mu)$,

$$d(\delta, 1 - \delta) \leq O(1) + \sum_{s,b} \mathbb{E}_\mu[N_{s,b}(\tau)] \text{KL}(\nu_{s,b,\varepsilon} \| \nu'_{s,b,\varepsilon}),$$

the $O(1)$ being the bounded boundary terms of (4.5). Writing $\nu(s, b) = \mathbb{E}_\mu[N_{s,b}(\tau)] / \mathbb{E}_\mu[\tau - R + 1] \in \Sigma([K]^{R+1})$ and proceeding as before,

$$d(\delta, 1 - \delta) \leq O(1) + \mathbb{E}_\mu[\tau - R + 1] T_{\varepsilon,R}^*(\mu).$$

Dividing by $\log(1/\delta)$, the $O(1)$ term vanishes and $d(\delta, 1 - \delta) / \log(1/\delta) \rightarrow 1$, yielding the bound. $\square$

Remark 4.2 (Always-repeat fallback). Letting $R \rightarrow \infty$ removes the uniform-reset term and gives the always-repeat strategy, whose characteristic time $T_\varepsilon^*(\mu)$ is the analogous supremum over distributions on $\bigcup_{r \geq 0} \{r\} \times [K]^{r+1}$ (the run length of consecutive erasures together with the corresponding window). The three strategies are ordered: always-repeat is the most informative and uniform the least, with R -tolerance interpolating between them.

4.3 Achievability: A Tracking-Based Algorithm

We now give a policy whose stopping time attains the bound of Proposition 4.3. Let ν_μ^* denote a maximizer of $T_{\varepsilon,R}^*(\mu)$.

4.3.1 Ergodicity and Estimation

Lemma 4.4 (Ergodicity). *Under the uniform arm-selection rule $A_n \sim \text{Unif}([K])$, the window chain $\{s(n)\}_{n \geq R}$ is an ergodic Markov chain on $[K]^R$ with a unique stationary distribution Φ_{unif} .*

Proof. Under the uniform rule every action has probability $1/K > 0$ at every state, and the shift kernel (3.3) can reach any window $(b_1, \dots, b_R)$ from any state in R steps by transmitting $b_1, \dots, b_R$ in order. Hence the chain is irreducible and aperiodic on the finite set $[K]^R$, and therefore ergodic with a unique stationary distribution. $\square$

Set $\nu_{\text{unif}}(s, b) = \frac{1}{K} \Phi_{\text{unif}}(s)$. For $(s, b) \in [K]^{R+1}$ define the empirical mean of the rewards collected at (s, b) ,

$$\hat{\mu}_{s,b}(n) = \frac{1}{N_{s,b}(n)} \sum_{t=R}^n \mathbf{1}\{s(t) = s, A_t = b\} X_t, \quad (4.6)$$

and let $\hat{\mu}_b(n) = K^{-R} \sum_s \hat{\mu}_{s,b}(n)$.

Lemma 4.5 (Consistency). *If $\min_{s,b} N_{s,b}(n) = \Omega(n^\alpha)$ for some $\alpha > 0$, then $\hat{\mu}_{s,b}(n) \rightarrow \mu_{s,b,\varepsilon}$ almost surely for all (s, b) , and $\hat{\mu}_b(n) \rightarrow \mu_b^\varepsilon$ almost surely.*

Proof sketch. Fix (s, b) . The centred summands $M_t = (X_t - \mu_{s,b,\varepsilon}) \mathbf{1}\{s(t) = s, A_t = b\}$ form a martingale-difference sequence with respect to the natural filtration, since by (4.4) the conditional mean of X_t given $\{s(t) = s, A_t = b\}$ is exactly $\mu_{s,b,\varepsilon}$. The conditional variances are bounded, so a strong law for martingales [LS20] gives $\sum_{t \leq n} M_t / N_{s,b}(n) \rightarrow 0$ a.s. whenever $N_{s,b}(n) \rightarrow \infty$, which holds under the stated growth condition. Averaging over s and using $K^{-R} \sum_s \mu_{s,b,\varepsilon} = \mu_b^\varepsilon$ gives the second claim. The full argument is in Appendix A.4. $\square$

4.3.2 Sampling, Stopping, and the Achievability Theorem

Let $\hat{\mu}(n) = [\hat{\mu}_b(n)]_b$ and

$$\nu_{n-1}^* = \arg \sup_{\nu \in \Sigma([K]^{R+1})} \inf_{\lambda \in \text{Alt}(\hat{\mu}(n-1))} \sum_{s,b} \nu(s, b) \text{KL}(\nu_{s,b,\varepsilon} \| \nu'_{s,b,\varepsilon}). \quad (4.7)$$

Fix a mixing parameter $\eta \in (0, 1)$ and select $A_n \sim \Phi_n(\cdot \mid s(n))$, where

$$\Phi_n(b \mid s) = \frac{\eta \nu_{\text{unif}}(s, b) + (1 - \eta) \nu_{n-1}^*(s, b)}{\eta \Phi_{\text{unif}}(s) + (1 - \eta) \sum_{b'} \nu_{n-1}^*(s, b')}. \quad (4.8)$$

The mixture guarantees forced exploration, $\Phi_n(b \mid s) \geq \frac{\eta}{K} \Phi_{\text{unif}}(s) > 0$, so the hypothesis of Lemma 4.5 holds. With the GLR statistic

$$Z(n) = \inf_{\lambda \in \text{Alt}(\hat{\mu}(n))} \sum_{s,b} N_{s,b}(n) \text{KL}(\hat{\nu}_{s,b,\varepsilon} \| \nu'_{s,b,\varepsilon}) \quad (4.9)$$

and threshold $\beta_R(n, \delta) = \log(C_{K,R} n^{1+\alpha}/\delta)$, the learner stops at

$$\tau = \inf \left\{ n \geq R : Z(n) \geq \beta_R(n, \delta), \min_{s,b} N_{s,b}(n) > 0 \right\} \quad (4.10)$$

and recommends $\hat{a} = \arg \max_b \hat{\mu}_b(\tau)$.

Theorem 4.6 (Achievability). *The policy $((4.8), (4.10), \hat{a})$ is δ -PC, and for every instance μ with a unique best arm,*

$$\limsup_{\delta \downarrow 0} \frac{\mathbb{E}_\mu[\tau]}{\log(1/\delta)} \leq \frac{1}{\eta T_{\text{unif}}^*(\mu) + (1 - \eta) T_{\varepsilon,R}^*(\mu)},$$

where $T_{\text{unif}}^*(\mu) = \inf_{\lambda \in \text{Alt}(\mu)} \sum_{s,b} \nu_{\text{unif}}(s,b) \text{KL}(\nu_{s,b,\varepsilon} \| \nu'_{s,b,\varepsilon})$. Letting $\eta \downarrow 0$, the bound matches Proposition 4.3, so the policy is asymptotically optimal.

Proof. Correctness. The choice of threshold $\beta_R(n, \delta)$ controls the probability that the GLR statistic ever exceeds it on a wrong recommendation: a method-of-mixtures / mixture-martingale bound [KK21] shows that $\mathbb{P}_\mu(\tau < \infty, \hat{a} \neq a^*(\mu)) \leq \delta$ for a constant $C_{K,R}$ depending only on K, R ; hence the policy is δ -PC.

Sample complexity. Forced exploration and Lemma 4.5 give $\hat{\mu}(n) \rightarrow \mu$ a.s., so four facts hold almost surely: (i) $\text{Alt}(\hat{\mu}(n)) = \text{Alt}(\mu)$ for all large n ; (ii) $\nu_n^* \rightarrow \nu_\mu^*$ by continuity of the maximizer in (4.7); (iii) by the ergodic theorem applied to the controlled chain under (4.8), $N_{s,b}(n)/n \rightarrow \eta \nu_{\text{unif}}(s,b) + (1 - \eta) \nu_\mu^*(s,b)$; (iv) consequently $Z(n)/n \rightarrow \eta T_{\text{unif}}^*(\mu) + (1 - \eta) T_{\varepsilon,R}^*(\mu)$. By the stopping rule, τ is of order $\beta_R(\tau, \delta)/[Z(\tau)/\tau]$, and since $\beta_R(n, \delta) = \log(1/\delta) + o(\log(1/\delta))$, fact (iv) yields the stated lim sup. As $\eta \downarrow 0$ the denominator tends to $T_{\varepsilon,R}^*(\mu)$, matching the converse. See Appendix A.5 for the omitted details. $\square$

4.4 The Implemented Algorithm E-NaS

Solving (4.7) exactly at every round is expensive, because the inner infimum over Alt ranges over a continuum and the mixture KLs have no closed form. For experimentation we use E-NAS (Algorithm 1), which exploits the MDP structure directly: it treats the K^R -state window chain with the deterministic shift kernel as a discounted best-policy-identification problem and applies the adaptive-sampling machinery of Al Marjani and Proutiere [AMP21]. This machinery assumes *bounded* rewards, which is why the experiments of Chapter 5 instantiate the arms as Bernoulli. Policy iteration produces value gaps that define hardness terms; a convex program over the

Algorithm 1 E-NAS: MDP reduction for finite-window erasure BAI

```
1: Input:  $\delta; K, R, \varepsilon$ ; discount  $\gamma$ ; constant  $\eta$ ; recompute cadence.  
2: Build  $\mathcal{S} = [K]^R$  and the deterministic kernel (3.3); set  $\hat{\mu}_{s,b} \leftarrow \frac{1}{2}$ ,  $N_{s,b} \leftarrow 0$ ,  
    $s(1) \sim \text{Unif}(\mathcal{S})$ .  
3: for  $t = 1, 2, \dots$  do  
4:    $(\hat{V}_t, \hat{Q}_t, \hat{\pi}_t) \leftarrow \text{POLICYITERATION}(P, \hat{\mu}, \gamma)$ ; gaps  $\hat{\Delta}_t(s, b) = \hat{V}_t(s) - \hat{Q}_t(s, b)$ .  
5:   Form hardness terms  $\hat{H}_{s,b}, \hat{H}^*$  and update the allocation  $\hat{\omega} \in \Omega$  by the convex  
   program.  
6:    $\eta_t \leftarrow 1/\sqrt{t}$ ; request  $A_t \sim$  the  $\eta_t$ -mixed allocation at  $s(t)$ ; observe  $X_t$ ; update  
    $N, \hat{\mu}$ ; shift to  $s(t+1)$ .  
7:    $U_t \leftarrow \max_{(s,b): b \neq \hat{\pi}_t(s)} \frac{\hat{H}_{s,b}}{N_{s,b}/t} + \frac{\hat{H}^*}{\min_s N_{s, \hat{\pi}_t(s)}/t}$ ;  $Z(t) \leftarrow t/U_t$ .  
8:   if  $\min_{s,b} N_{s,b} > 0$  and  $Z(t) \geq \beta(t, \delta)$  then  
9:     return  $\hat{a} = \text{majority vote of } \{\hat{\pi}_t(s) : s \in \mathcal{S}\}$ .  
10:  end if  
11: end for
```

shift-feasible occupancies Ω sets the allocation; forced exploration tracks it; and a GLR proxy with the threshold of (4.10) triggers stopping. The discount $\gamma \in (0, 1)$ is an algorithmic device that renders the value functions well defined.

4.5 The Uniform Case and the Ordering of Strategies

By Remark 4.1, uniform fallback reduces to classical BAI on the effective distributions $\nu_{a,\varepsilon,\text{unif}}$, whose means are μ^ε . When these rewards are bounded this has a useful practical consequence: a successive-elimination algorithm that maintains confidence intervals on the μ_a^ε and discards dominated arms is δ -PC and enjoys a *non-asymptotic* sample-complexity bound, obtained by the standard elimination analysis [EDMM06] applied to μ^ε . Concretely, ensuring that a sufficient fraction of each arm's transmissions are received unerased (which fails with probability at most $\varepsilon^{\Omega(t)}$ per round) and that the empirical means concentrate, a union bound delivers a δ -PC guarantee with the sample complexity scaling as $\sum_{a \neq a^*} 1/(\Delta_a^\varepsilon)^2$, where $\Delta_a^\varepsilon = \mu_{a^*}^\varepsilon - \mu_a^\varepsilon$.

Comparing the characteristic times confirms the intuition of Remark 4.2: among the three agent strategies, always repeating the previous arm minimizes the characteristic time and uniform sampling maximizes it, with R -tolerance interpolating monotonically as R grows. Repeating a successfully transmitted arm therefore wastes

the least information under erasures, making it the most effective fallback for best arm identification.

Chapter 5

Experiments and Evaluation

This chapter evaluates E-NAS empirically. We describe the experimental setup, then study correctness and how the empirical stopping time depends on the confidence level δ , the erasure probability ε , and the memory R . We close with computational observations.

5.1 Overview

The lower bound and the Track-and-Stop achievability result of the preceding chapters make no parametric assumption on the reward distributions. The tractable reduction E-NAS, however, builds on best-policy-identification machinery that assumes *bounded* rewards, so for a concrete and reproducible empirical study we instantiate the arms as Bernoulli. All experiments use E-NAS (Algorithm 1) on two-armed Bernoulli instances observed through the finite-memory erasure channel of Chapter 3. The aims are threefold: (i) to confirm that the GLR stopping rule delivers the δ -PC guarantee in practice; (ii) to verify the predicted $\log(1/\delta)$ scaling of the stopping time; and (iii) to quantify the price of erasures as ε and R vary.

5.2 Experimental Setup

5.2.1 Bandit Instance and Parameters

Unless stated otherwise, every experiment uses $K = 2$ Bernoulli arms with means $\mu = [0.30, 0.90]$ (so the true best arm is arm 2), forced-exploration constant $\eta = 1.0$, and discount $\gamma = 0.5$. Each swept configuration is run for 50 independent seeds.

Seeds are assigned deterministically from a per-experiment namespace so that runs are exactly reproducible.

5.2.2 Implementation Details

E-NAS recomputes its policy and allocation every 300 steps rather than at every round, which leaves the stopping time essentially unchanged while reducing wall-clock cost. The window MDP has K^R states; for the two-armed instance this is 2^R . Experiments were run on an Apple ARM64 machine with 11 logical CPUs and approximately 18 GiB of memory under Python 3.11.1; exact dependency and git-commit metadata are recorded with each run.

The complete implementation is openly available at <https://github.com/Deekshithrathod/Erasure-BAI>. It is written in Python and follows the reduction of Chapter 4 directly: it builds the length- R window MDP by enumerating the K^R window states and the deterministic shift kernel, evaluates the current candidate best policy by policy iteration on the discounted window MDP, fixes the sampling allocation by solving the convex best-policy-identification program, and stops via the generalized-likelihood-ratio test. Each experiment is launched from a deterministic seed namespace, and every run logs its resolved dependency versions together with the git commit that produced it, so that the results reported in this chapter can be regenerated end to end.

5.2.3 Evaluation Protocol

For each run we record whether the algorithm stopped, the identified arm, and the stopping time τ . A run is *correct* if it identifies arm 2. We report the empirical correct-identification rate, the mean and standard deviation of τ , and the mean per-run runtime.

5.3 Correctness

Across all completed configurations-1,100 runs in total-E-NAS stopped on every run and identified the correct arm every time, for an empirical error rate of 0%. Since the smallest confidence parameter in the correctness summary is $\delta = 0.1$, the observed error rate is comfortably below the target, indicating that the GLR threshold is conservative on these instances.

Table 5.1: Stopping time versus δ at $R = 3$, $\varepsilon = 0.3$ (50 runs each; all correct).

δ	$\mathbb{E}[\tau_\delta] \pm \text{sd}$
0.10	$120\,900 \pm 1\,873$
0.05	$122\,160 \pm 1\,421$
0.01	$125\,856 \pm 1\,926$

Table 5.2: Stopping time on the exponential grid $\delta = e^{-i}$ at $R = 2$, $\varepsilon = 0.3$ (50 runs each; all correct).

δ	$\log(1/\delta)$	$\mathbb{E}[\tau_\delta] \pm \text{sd}$	$\Delta\bar{\tau}_\delta$ per nat
e^{-1}	1	$29\,820 \pm 851$	–
e^{-2}	2	$30\,756 \pm 777$	+936
e^{-3}	3	$31\,770 \pm 662$	+1 014
e^{-4}	4	$32\,700 \pm 606$	+930
e^{-5}	5	$33\,552 \pm 775$	+852
e^{-6}	6	$34\,254 \pm 884$	+702
e^{-7}	7	$35\,148 \pm 993$	+894
e^{-8}	8	$35\,826 \pm 842$	+678
e^{-9}	9	$36\,636 \pm 777$	+810
e^{-10}	10	$37\,212 \pm 771$	+576
e^{-11}	11	$37\,860 \pm 688$	+648
e^{-12}	12	$38\,676 \pm 927$	+816
e^{-13}	13	$39\,246 \pm 909$	+570
e^{-14}	14	$39\,834 \pm 805$	+588
e^{-15}	15	$40\,566 \pm 1\,052$	+732

5.4 Effect of the Confidence Level

We first vary δ with the channel fixed. Table 5.1 reports the coarse sweep ($R = 3$, $\varepsilon = 0.3$), and Table 5.2 the fine exponential grid ($R = 2$, $\varepsilon = 0.3$) used to expose the scaling law.

The fine grid is the key result: as $\delta = e^{-i}$ ranges over $i = 1, \dots, 15$, the mean stopping time rises essentially linearly in $i = \log(1/\delta)$, from 29,820 to 40,566. This linear-in- $\log(1/\delta)$ growth is exactly the signature predicted by Proposition 4.3, with the slope estimating the characteristic time $T_{\varepsilon,R}^*(\mu)$ for this instance. The final column reports the per-nat increment $\Delta\bar{\tau}_\delta$; it stays in a narrow band around 700–900 samples, an essentially constant slope that is the empirical estimate of this characteristic time.

Table 5.3: Stopping time versus erasure probability ε at $R = 2$, $\delta = 0.1$ (50 runs each; all correct).

ε	$\mathbb{E}[\tau_\delta] \pm \text{sd}$	vs. $\varepsilon = 0.01$
0.01	$18\,840 \pm 388$	$\times 1.00$
0.05	$19\,908 \pm 524$	$\times 1.06$
0.10	$21\,480 \pm 588$	$\times 1.14$
0.30	$31\,074 \pm 820$	$\times 1.65$

Table 5.4: Stopping time versus memory R at $\varepsilon = 0.3$, $\delta = 0.1$ (50 runs each; all correct).

R	State space	State $\times$ Action space	$\mathbb{E}[\tau_\delta] \pm \text{sd}$
1	2	4	$10\,320 \pm 514$
2	4	8	$31\,194 \pm 690$
3	8	16	$120\,522 \pm 1\,558$
4	16	32	$479\,700 \pm 4\,289$
5	32	64	$1\,919\,136 \pm 10\,198$
6	64	128	$7\,715\,082 \pm 33\,591$

5.5 Effect of the Erasure Probability

Holding the instance and confidence fixed ($R = 2$, $\delta = 0.1$), we vary the erasure probability ε . Table 5.3 shows that the stopping time falls monotonically as ε decreases: from 31,074 samples at $\varepsilon = 0.3$ to 18,840 at $\varepsilon = 0.01$, a reduction of roughly 40%, equivalently a $\times 1.65$ inflation at $\varepsilon = 0.3$ relative to the near-clean $\varepsilon = 0.01$ baseline, as the final column shows. This directly quantifies the price of erasures: a noisier channel forces the learner to collect more samples to reach the same confidence, consistent with the dilution of per-request information discussed in Chapter 4.

5.6 Effect of the Memory Parameter

Finally we vary the memory $R \in \{1, 2, 3, 4, 5, 6\}$ with the channel and confidence fixed at $\varepsilon = 0.3$, $\delta = 0.1$. The window MDP has $K^R = 2^R$ states and $|\mathcal{S} \times \mathcal{A}| = 2^{R+1}$ state-action pairs, which grows from 4 at $R = 1$ to 128 at $R = 6$. Table 5.4 reports the resulting stopping times.

The stopping time grows *geometrically* in R : the ratio between successive rows settles at roughly $\times 4$ per unit increase of R , so a longer memory is markedly more expensive to identify through. This reflects both the exponential growth of the state

space and the greater mixing induced by a longer window, matching the R dependence anticipated in Chapter 4.

5.7 Computational Considerations

Two effects of the window size are visible. First, runtime per run grows sharply with R : at $R = 2$ a run takes around 1–2 seconds, whereas at $R = 3$ (Table 5.1) the larger state space and longer stopping times push this to roughly 19 seconds. Second, the stopping times themselves are far larger at $R = 3$ than at $R = 2$, reflecting both the increased number of state–action pairs that must be explored and the greater mixing induced by a longer memory. These observations motivate the $R \rightarrow \infty$ and approximation directions discussed in Chapter 6.

5.8 Summary

The experiments confirm the central predictions of the theory: E-NAS is reliably correct, its stopping time scales linearly in $\log(1/\delta)$, and the sample complexity increases with the erasure probability. The memory sweep completes the empirical picture, showing that the stopping time grows geometrically with R as the window MDP expands.

Chapter 6

Interpretation and Analysis

The lower bound and algorithms of Chapter 4 are operational but opaque: they say *what* the optimal sampling effort is without explaining *why* erasures are costly or why the resulting Markov chain behaves well enough for the guarantees to hold. This chapter supplies that interpretation.

6.1 Overview

We make three points. First, erasures act as *information dilution*: each request carries less evidence about the requested arm, and the characteristic time $T^*(\mu)$ measures exactly how much. Second, $T^*(\mu)$ is the value of a two-player game whose equilibrium explains the shape of the optimal allocation. Third, the controlled window chain is ergodic under forced exploration, which is the technical fact underpinning both the lower bound’s occupancy argument and the algorithms’ consistency.

6.2 Erasures as Information Dilution

Under clean feedback, transmitting arm b yields a sample of its own reward distribution ν_b , so each pull contributes full information about arm b . Under the channel, the same transmission yields a sample of the mixture $\nu_{s,b,\varepsilon}^\mu$ of (3.1), which places only weight $1 - \varepsilon$ on the transmitted arm and spreads the rest across the window and the uniform reset. The KL divergence appearing in the lower bound of Proposition 4.3—the rate at which a transmission distinguishes μ from an alternative λ —is therefore strictly smaller than its clean-feedback counterpart whenever $\varepsilon > 0$. Since $T_{\varepsilon,R}^*(\mu)$ is the inverse of this rate, optimized over occupancies, erasures inflate it. The empiri-

cal ε -sweep in Chapter 5-stopping time rising from 18,840 at $\varepsilon = 0.01$ to 31,074 at $\varepsilon = 0.3$ -is the visible footprint of this dilution.

6.3 The Characteristic Time as a Two-Player Game

The characteristic time $T_{\varepsilon,R}^*(\mu)$ of Proposition 4.3 is a max–min, hence a zero-sum game. The *learner* chooses a state–action occupancy $\nu \in \Sigma([K]^{R+1})$; an adversary, *Nature*, chooses the most confusing alternative $\lambda \in \text{Alt}(\mu)$. The objective $\sum_{s,b} \nu(s,b) \text{KL}(\cdot)$ is linear in ν and the inner infimum over λ makes it concave in ν ; since $\Sigma([K]^{R+1})$ is compact and convex, a maximizing occupancy ν_μ^* exists. At equilibrium the learner spreads its transmissions so that no single alternative is cheap to leave unrefuted, while Nature concentrates on the hardest alternative—typically the one that swaps the best arm with its closest competitor. The channel reshapes this game: because evidence about an arm now arrives partly through transmissions of *other* arms (via the window), the equilibrium occupancy differs from the clean-feedback one.

6.4 Ergodicity of the Controlled Window Chain

Both the lower bound (Proposition 4.3) and the achievability theorem (Theorem 4.6) rely on the empirical state–action occupancy converging to a well-defined limit. This is exactly the content of the ergodicity and consistency results of Chapter 4: under the forced-exploration sampling rule every action has positive probability at every state, so by Lemma 4.4 the controlled window chain is ergodic with a unique stationary distribution, and by Lemma 4.5 the arm-mean estimates are consistent. Together these make the empirical visit counts concentrate around the tracked occupancy, which is what closes both arguments.

6.5 Qualitative Analysis of the Allocation

Inspecting the allocations produced by E-NAS clarifies its behaviour. The algorithm does not request arms uniformly: it over-samples the states and actions that the hardness terms flag as most informative for separating the empirical best arm from its competitor, while forced exploration guarantees the remaining state–action pairs are not starved. The larger stopping times at $R = 3$ versus $R = 2$ reflect that a longer memory both enlarges the set of state–action pairs that must each be visited

and lengthens the chain’s mixing, so the same total evidence is accumulated more slowly.

6.6 Connection to Classical BAI and MDP Identification

Two sanity checks place the results in context. As $\varepsilon \rightarrow 0$ the mixtures degenerate to the single arm distributions ν_a , the window states decouple, and the framework reduces to standard Track-and-Stop with characteristic time $T_{\text{clean}}^*(\mu)$ [GK16]. As the memory grows, the problem approaches the MDP best-policy-identification problem of [AMP21], of which E-NAS is a direct instantiation. The erasure-BAI problem thus sits on a continuum between these two well-understood endpoints, and the parameters (ε, R) are the coordinates that locate it.

6.7 Summary

Erasures cost samples because they dilute per-transmission information; the optimal response is the equilibrium of a learner–Nature game over state–action occupancies; and forced exploration makes the underlying chain ergodic, which is what makes the whole analysis go through. These insights motivate the open directions collected in the next chapter.

Chapter 7

Conclusion and Future Work

7.1 Summary

This thesis studied best arm identification in the fixed-confidence regime when feedback reaches the learner through a finite-memory erasure channel served by a resource-constrained agent. We argued that this setting is the natural model for identification problems in which a low-power device acts in the environment and reports over an unreliable link, and we showed that the single memory parameter R unifies the uniform-reset and always-repeat behaviours that earlier work treated separately. Casting the agent’s length- R request window as the state of a controlled Markov chain, we derived an information-theoretic lower bound on the expected sample complexity of any δ -probably-correct policy, expressed as a max–min program over state–action occupancies. We showed the bound is matched, in the small- δ limit, by a mixture-KL Track-and-Stop procedure, and we proposed E-NAS, a tractable reduction to discounted best-policy identification on the window MDP. Experiments on Bernoulli instances confirmed reliable identification, the predicted linear-in- $\log(1/\delta)$ scaling of the stopping time, and a clear increase in sample complexity with the erasure probability—a direct, quantitative measurement of the price of erasures.

7.2 Future Work

Erasure feedback to the learner. We assumed the learner never observes whether an erasure occurred. Granting it the erasure indicator (but not the realized arm) is a mild and realistic relaxation that should reduce the characteristic time; quantifying that reduction is a natural next step.

The infinite-memory limit. As $R \rightarrow \infty$ the channel always repeats on an erasure and the window MDP has K^R states, so neither the Track-and-Stop target nor E-NAS scales directly. Characterizing $T^*(\mu)$ in this limit, and designing approximations whose cost does not grow as K^R , would make the framework practical for long memories.

Non-asymptotic guarantees. Our optimality statements are asymptotic in δ . The empirical conservatism of the GLR threshold (zero errors well within the target) suggests room for sharper, finite-confidence bounds on both the error probability and the stopping time.

Adaptivity to unknown channel parameters. We assumed ε and R are known. Estimating them online, or designing policies robust to their misspecification, is important for deployment, since a real channel rarely advertises its statistics.

Broader extensions. Finally, the framework invites extension beyond the present scope: empirical validation beyond the Bernoulli instances studied here (the lower bound and the Track-and-Stop target already cover general distributions, while E-NAS requires bounded rewards), correlated rather than independent erasures, more than two arms at larger R , and the fixed-budget counterpart of the problem. Each would test how far the mixture-KL and MDP-reduction viewpoints developed here continue to apply.

Appendix A

Proofs and Derivations

This appendix collects the derivations and full proofs deferred from Chapters 3 and 4. Throughout, ν_a denotes the reward distribution of arm a and μ_a its mean; A_t is the arm transmitted by the learner at time t and $\tilde{A}_t$ the arm the agent pulls; and $\mathcal{E}_t$ is the event that an erasure occurs at time t , so that $\mathbb{P}(\mathcal{E}_t) = \varepsilon$ independently across t .

A.1 Derivation of the Effective Reward Law

We derive the conditional law of the observed reward underlying the mixture (3.1) and the conditional density (4.4). Fix $t \geq 0$ and condition on the history $(A_{0:t}, X_{0:t-1})$. Splitting on the erasure event at time t , for every measurable $G \subseteq \mathbb{R}$,

$$\mathbb{P}_\mu(X_t \in G \mid A_{0:t}, X_{0:t-1}) = (1 - \varepsilon) \mathbb{P}_\mu(X_t \in G \mid \tilde{A}_t = A_t) + \varepsilon \mathbb{P}_\mu(X_t \in G \mid \tilde{A}_t = \tilde{A}_{t-1}), \quad (\text{A.1})$$

since with probability $1 - \varepsilon$ the command arrives and the agent pulls A_t , and with probability ε it is erased and the agent repeats the previously pulled arm $\tilde{A}_{t-1}$. Under the R -tolerant rule the agent repeats for at most R consecutive losses, so applying (A.1) recursively to $\tilde{A}_{t-1}, \tilde{A}_{t-2}, \dots$ and resetting to a uniform pull on the $(R+1)$ st consecutive erasure gives, for $t \geq R$,

$$\mathbb{P}_\mu(X_t \in G \mid A_{0:t}, X_{0:t-1}) = (1 - \varepsilon) \sum_{s=0}^R \varepsilon^s \nu_{A_{t-s}}(G) + \frac{\varepsilon^{R+1}}{K} \sum_{a'=1}^K \nu_{a'}(G). \quad (\text{A.2})$$

Passing to densities recovers (4.4); collecting the last R transmitted arms into the window $s = (A_{t-R}, \dots, A_{t-1})$ and writing $b = A_t$ recovers the mixture (3.1). Taking

expectations in (A.2),

$$\mathbb{E}_\mu[X_t \mid A_{0:t}, X_{0:t-1}] = (1 - \varepsilon) \sum_{s=0}^R \varepsilon^s \mu_{A_{t-s}} + \frac{\varepsilon^{R+1}}{K} \sum_{a'=1}^K \mu_{a'}, \quad (\text{A.3})$$

the conditional mean quoted in Section 3.2.4. For the always-repeat rule the recursion never resets (except through the convention $\tilde{A}_0 \sim \text{Unif}([K])$ on an initial erasure), and unrolling all the way to $s = t$ yields the full-history law

$$\mathbb{P}_\mu(X_t \in G \mid A_{0:t}, X_{0:t-1}) = (1 - \varepsilon) \sum_{u=0}^t \varepsilon^{t-u} \nu_{A_u}(G) + \frac{\varepsilon^{t+1}}{K} \sum_{a'=1}^K \nu_{a'}(G), \quad (\text{A.4})$$

with conditional mean $(1 - \varepsilon) \sum_{u=0}^t \varepsilon^{t-u} \mu_{A_u} + \frac{\varepsilon^{t+1}}{K} \sum_{a'} \mu_{a'}$, a function of the *entire* selection history $A_{0:t}$.

A.2 The Expected Log-Likelihood Ratio

From the definition (4.1), the arm-selection factors cancel because the sampling rule is oblivious to the instance, leaving

$$Z_{\mu,\lambda}(n) = \sum_{t=0}^n \log \frac{\mathbb{P}_\mu(X_t \mid A_{0:t}, X_{0:t-1})}{\mathbb{P}_\lambda(X_t \mid A_{0:t}, X_{0:t-1})}.$$

Taking $\mathbb{E}_\mu$ and applying the tower property with the conditional law (A.2), the inner conditional expectation of each summand is the Kullback–Leibler divergence between the effective distributions, so

$$\mathbb{E}_\mu[Z_{\mu,\lambda}(n)] = \mathbb{E}_\mu \left[\sum_{t=0}^n \text{KL}(\nu_{s(t), A_t, \varepsilon} \parallel \nu'_{s(t), A_t, \varepsilon}) \right],$$

where for $t < R$ the window $s(t)$ is truncated. Splitting the sum at $t = R$ and writing $N_{s,b}(n) = \sum_{t=R}^n \mathbf{1}\{s(t) = s, A_t = b\}$,

$$\mathbb{E}_\mu[Z_{\mu,\lambda}(n)] = O(1) + \sum_{s \in [K]^R} \sum_{b=1}^K \mathbb{E}_\mu[N_{s,b}(n)] \text{KL}(\nu_{s,b,\varepsilon} \parallel \nu'_{s,b,\varepsilon}),$$

the $O(1)$ collecting the R boundary terms, each a divergence between fixed distributions and hence bounded uniformly in n . This is (4.5).

A.3 Proof of the Transportation Inequality (Lemma 4.1)

Fix $\pi \in \Pi(\delta)$ with $\mathbb{E}_\mu[\tau] < \infty$ (otherwise the bound is vacuous) and $\lambda \in \text{Alt}(\mu)$. Consider the event $E = \{\hat{a} = a^*(\mu)\} \in \mathcal{F}_\tau$. Because π is δ -PC, $\mathbb{P}_\mu(E) \geq 1 - \delta$; and since $a^*(\lambda) \neq a^*(\mu)$, declaring $a^*(\mu)$ is an error under λ , so $\mathbb{P}_\lambda(E) \leq \delta$. By Wald's identity the expected log-likelihood ratio at the stopping time equals the relative entropy of the two laws restricted to $\mathcal{F}_\tau$, and the data-processing inequality for the binary statistic $\mathbf{1}_E$ gives

$$\mathbb{E}_\mu[Z_{\mu,\lambda}(\tau)] = \text{KL}(\mathbb{P}_\mu^{\mathcal{F}_\tau} \parallel \mathbb{P}_\lambda^{\mathcal{F}_\tau}) \geq d(\mathbb{P}_\mu(E), \mathbb{P}_\lambda(E)) \geq d(\delta, 1 - \delta),$$

the last step using the monotonicity of d together with $\mathbb{P}_\mu(E) \geq 1 - \delta \geq \mathbb{P}_\lambda(E)$; see also [KCG16, Lemma 19]. Combined with the decomposition of Appendix A.2, this is the inequality used in the lower bounds of Section 4.2. Finally $d(\delta, 1 - \delta) \geq \log(1/(2.4\delta))$, so $d(\delta, 1 - \delta)/\log(1/\delta) \rightarrow 1$ as $\delta \downarrow 0$.

A.4 Proof of the Consistency Lemma (Lemma 4.5)

Fix $(s, b) \in [K]^{R+1}$ and let $\mathcal{F}_t = \sigma(A_{0:t}, X_{0:t})$. Define

$$M_t = \mathbf{1}\{s(t) = s, A_t = b\} (X_t - \mu_{s,b,\varepsilon}).$$

By (A.2) the conditional mean of X_t on the event $\{s(t) = s, A_t = b\}$ is exactly $\mu_{s,b,\varepsilon}$, the mean of $\nu_{s,b,\varepsilon}$; hence $\mathbb{E}_\mu[M_t \mid \mathcal{F}_{t-1}, A_t] = 0$ and $\{M_t\}$ is a martingale-difference sequence adapted to $\{\mathcal{F}_t\}$. Under the standing assumption that the rewards have uniformly bounded conditional variance (in particular for the bounded rewards of Chapter 5), the strong law of large numbers for martingale differences [LS20] yields

$$\frac{1}{N_{s,b}(n)} \sum_{t=R}^n M_t \xrightarrow{n \rightarrow \infty} 0 \quad \text{almost surely on } \{N_{s,b}(n) \rightarrow \infty\}.$$

The hypothesis $\min_{s,b} N_{s,b}(n) = \Omega(n^\alpha)$ ensures $N_{s,b}(n) \rightarrow \infty$, so $\hat{\mu}_{s,b}(n) \rightarrow \mu_{s,b,\varepsilon}$ a.s. For the second claim, average over $s \in [K]^R$: a direct computation from (A.3), using that each window receives equal weight K^{-R} and that a uniformly random window makes each lagged arm uniform on $[K]$, gives $K^{-R} \sum_s \mu_{s,b,\varepsilon} = (1 - \varepsilon)\mu_b + \frac{\varepsilon}{K} \sum_{a'} \mu_{a'} = \mu_b^\varepsilon$, whence $\hat{\mu}_b(n) \rightarrow \mu_b^\varepsilon$ a.s.

A.5 Proof of the Achievability Theorem (Theorem 4.6)

Correctness. With the threshold $\beta_R(n, \delta) = \log(C_{K,R} n^{1+\alpha}/\delta)$, a method-of-mixtures (peeling) bound on the generalized-likelihood-ratio statistic $Z(n)$ of (4.9), together with a union bound over the finitely many candidate wrong recommendations, controls the probability that Z ever crosses the threshold in favour of an incorrect arm: $\mathbb{P}_\mu(\tau < \infty, \hat{a} \neq a^*(\mu)) \leq \delta$ for a constant $C_{K,R}$ depending only on K and R [KK21]. Hence the policy is δ -PC.

Sample complexity. The selection rule (4.8) keeps $\Phi_n(b \mid s) \geq \frac{\eta}{K} \Phi_{\text{unif}}(s) > 0$, so every state-action pair is visited at a rate bounded away from zero and $\min_{s,b} N_{s,b}(n) = \Omega(n)$; by Lemma 4.5, $\hat{\mu}(n) \rightarrow \mu$ almost surely. Four consequences then hold almost surely:

1. $\text{Alt}(\hat{\mu}(n)) = \text{Alt}(\mu)$ for all large n , since $a^*(\mu)$ is unique and $\hat{\mu}(n) \rightarrow \mu$;
2. $\nu_n^* \rightarrow \nu_\mu^*$, by continuity of the maximizer in (4.7) (Berge's maximum theorem, the inner infimum being continuous in the estimate);
3. by the ergodic theorem for the controlled window chain under the η -mixed rule (4.8), which is ergodic by Lemma 4.4, $N_{s,b}(n)/n \rightarrow \eta \nu_{\text{unif}}(s, b) + (1 - \eta) \nu_\mu^*(s, b)$;
4. consequently $Z(n)/n \rightarrow \eta T_{\text{unif}}^*(\mu) + (1 - \eta) T_{\varepsilon,R}^*(\mu) =: g(\eta)$.

The stopping rule gives $\tau \approx \beta_R(\tau, \delta)/(Z(\tau)/\tau)$, and since $\beta_R(n, \delta) = \log(1/\delta) + o(\log(1/\delta))$, dividing by $\log(1/\delta)$ and using item (iv) yields $\limsup_{\delta \downarrow 0} \mathbb{E}_\mu[\tau]/\log(1/\delta) \leq 1/g(\eta)$. Letting $\eta \downarrow 0$, $g(\eta) \rightarrow T_{\varepsilon,R}^*(\mu)$, which matches the converse of Proposition 4.3; hence the policy is asymptotically optimal.

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